

Set 4: Application of Le Chatelier's Principle

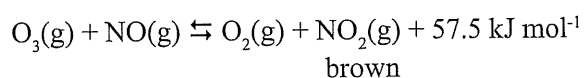
Le Chatelier was a French industrial chemist who spent a great deal of his life investigating equilibrium systems. He found that in many cases, a system at equilibrium would react in a predictable way when a change was imposed on the system. This ability to predict the impact of a change imposed on a system is called Le Chatelier's Principle:

When a change is imposed on a system at equilibrium, the system will move in such a way as to partially counteract that change.

Le Chatelier's Principle can be used to predict the effect of changing temperature, pressure and concentration on a system at equilibrium. Adding a catalyst has no effect on equilibrium yield as it increases the rate of the forward and reverse reactions equally.

Example

A reaction that is significant in the formation of smog is one between ozone and nitrogen monoxide. It can be represented by the following equation.



- (a) Write the equilibrium constant expression for this equation.
- (b) Predict the effect on the yield of NO_2 due to the following changes being made to the system at equilibrium.
- (i) More O_3 is added to the system (a by-product of the combustion of petrol)
 - (ii) NO_2 is removed from the system.
 - (iii) The volume of the system is increased.
 - (iv) The temperature of the system is decreased.

(a) $K = \frac{[\text{O}_2][\text{NO}_2]}{[\text{O}_3][\text{NO}]}$

- (b)(i) Increase in yield, $[\text{NO}_2]$
(ii) Increase in yield, $[\text{NO}_2]$
(iii) no change (the number of gas particles is the same on both sides of the equation)
(iv) increase in yield, $[\text{NO}_2]$ (the forward reaction is exothermic)

Set 4: Exercises

1. The industrial production of ammonia involves a reaction between nitrogen and hydrogen in the presence of an iron/iron oxide catalyst and can be represented by the following equation:



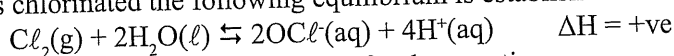
- (a) Write an equilibrium constant expression for the equation.
- (b) Predict the effect on the concentration of ammonia due to the following changes being made to the above reaction at equilibrium.
- (i) Increase the temperature,
 - (ii) Inject more $\text{N}_2(\text{g})$ into the system,
 - (iii) Reduce the volume of the system,
 - (iv) Remove the catalyst.

Notes

Set 4: Application of Le Chatelier's Principle

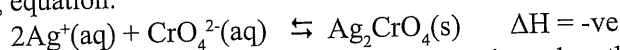
Notes

2. When water is chlorinated the following equilibrium is established:



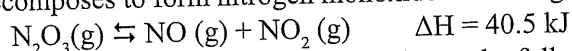
- (a) Write an equilibrium constant expression for the equation.
(b) Predict the effect on the concentration of $\text{OCl}^-(\text{aq})$ due to the following changes:
(i) More $\text{Cl}_2(\text{g})$ is added to the system,
(ii) The temperature is decreased,
(iii) A small amount of concentrated NaOH solution is added to the system,
(iv) A small amount of concentrated HCl solution is added to the system.

3. Ag_2CrO_4 is a slightly soluble salt and forms a saturated solution. This can be represented by the following equation:



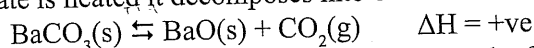
- (a) Predict the initial effect on the rate of the forward reaction when the equilibrium is re-established after the following changes have been applied:
(i) A few drops of concentrated NaCl solution is added to the system.
(ii) The solution is diluted by the addition of a small amount of water.
(iii) The temperature is increased.
(b) Predict the effect on the concentration of Ag^+ when equilibrium is re-established after the following changes have been applied:
(i) A few drops of concentrated NaCl solution is added to the system.
(ii) Some solid Ag_2CrO_4 is added to the system.
(iii) The temperature is increased.

4. Dinitrogen trioxide decomposes to form nitrogen monoxide and nitrogen dioxide.



- a) Predict the effect on the rate of the forward reaction due to the following changes.
(i) Extra N_2O_3 is pumped into the system at constant volume.
(ii) The temperature is increased.
(iii) The volume of the system is doubled.
b) Predict the effect on the concentration of products when equilibrium is re-established when the following changes are applied.
(i) Extra N_2O_3 is pumped into the system at constant volume.
(ii) The temperature is decreased.
(iii) The volume of the system is doubled.

5. When barium carbonate is heated it decomposes into barium oxide and carbon dioxide.



- a) Predict the effect on the rate of the forward reaction due to the following changes.
(i) The BaCO_3 is ground up more finely.
(ii) The system is cooled.
(iii) CO_2 is removed.
b) Predict the effect on the concentration of carbon dioxide when equilibrium is re-established due to the following changes.
(i) More BaCO_3 is added.
(ii) The volume is doubled.
(iii) The temperature is increased.

ANSWERS

Properties and Reactions

Set 1: Gases

- (i) 288 K (ii) 390 K (iii) 168 K
 - (b) (i) 25 °C (ii) 200 °C (iii) -161 °C
- (a) 2.17×10^{-2} mol (b) 3.38×10^{-2} mol (c) 1.46 mol
- (a) 93.7 g mol^{-1} (b) 11.0 g mol^{-1} (c) 92.7 g mol^{-1}
- (a) 28.8 g mol^{-1} (b) C_2H_4 (ethene)
- 0.108 g
- (a) (i) $\text{X} + 2\text{H}^+ \rightarrow \text{X}^{2+} + \text{H}_2$ (ii) $2\text{X} + 6\text{H}^+ \rightarrow 2\text{X}^{3+} + 3\text{H}_2$
 - (b) (i) 22.7 g mol^{-1} (ii) 45.4 g mol^{-1} (iii) 68.1 g mol^{-1}
 - (c) Sodium, as closest molar mass is Na.

Set 2: Limiting Reagents

- (a) NaHCO_3 (b) $8.51 \times 10^{-2} \text{ L}$
- (a) CO (b) 1.00 L (c) 1.00 L CO_2 , 0.500 L O_2 , no CO
- (a) $\text{V}(\text{CO}_2) = 4.00 \times 10^{-2} \text{ L}$, $\text{V}(\text{H}_2\text{O}) = 4.00 \times 10^{-2} \text{ L}$, $\text{V}(\text{O}_2) = 2.00 \times 10^{-2} \text{ L}$, $\text{V}(\text{C}_2\text{H}_4) = 0$
 - (b) 0.100 L
- (a) Cu (b) 1.02 L
- 238 kPa
- 16.0 L
- 133 kPa
- $6.01 \times 10^5 \text{ g}$ or 0.601 tonne
- (a) KNO_3 (b) 15.0 g (c) $1.81 \times 10^3 \text{ kPa}$

Set 3: Equilibrium constant expressions

- $$K = \frac{[\text{H}_2\text{O}]^2}{[\text{H}_2]^2[\text{O}_2]}$$
- $$K = \frac{[\text{N}_2]^2[\text{O}_2]}{[\text{N}_2\text{O}]^2}$$
- $$K = \frac{[\text{CH}_3\text{OH}]}{[\text{CO}][\text{H}_2]^2}$$
- $$K = [\text{Ag}^+]^2[\text{CrO}_4^{2-}]$$
- $$K = \frac{[\text{HCO}_3^-][\text{H}^+]}{[\text{CO}_2]}$$
- $$K = \frac{[\text{SO}_4^{2-}][\text{H}_3\text{O}^+]}{[\text{HSO}_4^-]}$$
- $$K = \frac{[\text{CrO}_4^{2-}]^2[\text{H}^+]^2}{[\text{Cr}_2\text{O}_7^{2-}]}$$
- $$K = [\text{CO}_2]$$
- $$K = \frac{[\text{CO}_3^{2-}][\text{NH}_4^+]}{[\text{HCO}_3^-][\text{NH}_3]}$$
- $$K = \frac{1}{[\text{SO}_3]}$$
- $$K = [\text{H}_2\text{O}][\text{CO}_2]$$
- $$K = \frac{1}{[\text{Cl}_2]}$$
- $$K = \frac{[\text{Ca}^{2+}][\text{HCO}_3^-]^2}{[\text{CO}_2]}$$
- $$K = \frac{[\text{CO}_3^{2-}][\text{NH}_4^+]}{[\text{HCO}_3^-][\text{NH}_3]}$$

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- (a) $K = \frac{[\text{NH}_3]^2}{[\text{N}_2][\text{H}_2]^3}$
 - (b) (i) decrease (ii) increase (iii) increase (iv) no change

2. (a) $K = \frac{[\text{OCl}^-]^2 [\text{H}^+]^4}{[\text{Cl}_2]}$
- (b) (i) increase (ii) increase (iii) increase (iv) decrease
3. (a) (i) decrease (ii) decrease (iii) increase
- (b) (i) decrease (ii) increase (iii) increase
4. (a) (i) increase (ii) increase (iii) decrease
- (b) (i) increase (ii) increase (iii) increase
5. (a) (i) increase (ii) decrease (iii) decrease
- (b) (i) no change (ii) no change (iii) increase

Set 5: Percentage composition and yield

1. (a) 69.9% (b) 92.9%
2. 76.8%
3. (a) 121 g (b) 77.3%
4. (a) $\text{Ca}(\text{OH})_2 \rightarrow \text{CaO} + \text{H}_2\text{O}$ (b) 76.4%
5. (a) $\text{Ba}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{BaSO}_4(\text{s})$ (b) 0.926 g (c) 45.5%
6. 82.5%
7. 74.9%

Set 6: Empirical formulas

1. $\text{Cu}_3(\text{CO}_3)_2(\text{OH})_2$
2. (a) CH_2
- (b) MFM = 56.00 g mol⁻¹, C_4H_8
3. $\text{Pb}_3\text{O}_2\text{Cl}_2$
4. (a) $\text{C}_2\text{H}_6\text{O}$ (b) MFM = 46.0 g mol⁻¹, $\text{C}_2\text{H}_6\text{O}$
5. CoC_4O_4 (b) MFM = 59.0 g mol⁻¹, $\text{C}_3\text{H}_9\text{N}$
6. (a) $\text{C}_3\text{H}_9\text{N}$ (b) $\text{C}_6\text{H}_4\text{Cl}_2$
7. (a) $\text{C}_3\text{H}_2\text{Cl}$ (b) $\text{C}_6\text{H}_7\text{N}$
8. (a) $\text{C}_6\text{H}_7\text{N}$
9. $\text{K}_2\text{Ni}(\text{C}_2\text{O}_4)_2 \cdot 2\text{H}_2\text{O}$
10. $\text{S}_4\text{Cl}_4\text{O}_{17}$

Set 7: Reaction types (more than one reaction may occur)

1. (a) $\text{CH}_3\text{COOH}(\ell) + \text{H}_2\text{O}(\ell) \rightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
- (b) $\text{NH}_3(\text{g}) + \text{H}_2\text{O}(\ell) \rightarrow \text{NH}_4^+(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
- $\text{NaHCO}_3(\text{s}) \rightarrow \text{Na}^+(\text{aq}) + \text{HCO}_3^-(\text{aq})$
- (c) $\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightarrow \text{H}_2\text{CO}_3(\text{aq}) + \text{OH}^-(\text{aq})$
- $\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightarrow \text{CO}_3^{2-}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
- (d) $\text{NaHSO}_4(\text{aq}) \rightarrow \text{Na}^+(\text{aq}) + \text{HSO}_4^-(\text{aq})$
- $\text{HSO}_4^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightarrow \text{SO}_4^{2-}(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
- (e) $\text{K}_2\text{CO}_3(\text{s}) \rightarrow 2\text{K}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$
- $\text{CO}_3^{2-}(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{OH}^-(\text{aq})$
- (f) $\text{NH}_4\text{CH}_3\text{COO}(\text{s}) \rightarrow \text{NH}_4^+(\text{aq}) + \text{CH}_3\text{COO}^-(\text{aq})$
- $\text{NH}_4^+(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{NH}_3(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
- $\text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{CH}_3\text{COOH}(\text{aq}) + \text{OH}^-(\text{aq})$